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PROJECT ON

BLINKIT GROCERY SALES & OUTLET ANALYSIS

Using Power BI, Power Query & DAX

Presented by
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About me

- I have completed my M.Sc. in Zoology and I am interested in learning Data Science because I want to develop my analytical and technical skills and explore how data can be used to understand patterns, solve problems, and support better decision making.

- I currently have no professional work experience and am building my skills through learning and practical projects.

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Agenda

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INTRODUCTION

- BlinkIT is an online grocery delivery platform offering a wide range of grocery and household products.
- This project analyzes grocery sales and outlet performance using Power BI.
- The analysis focuses on products, sales, outlet types, outlet sizes, locations, and establishment years.
- An interactive Power BI dashboard was developed to identify sales patterns, performance trends, and business opportunities.

Business Problem Statement

Blinkit management wants to understand and improve overall business performance by analyzing -

- Product Performance – Identify high-performing and underperforming products.
- Outlet Performance – Compare sales across different outlet types, sizes, and locations.
- Sales Trends – Analyze sales patterns and identify key factors affecting revenue.
- Customer Satisfaction – Understand how customer ratings vary across products and outlets.
- Visibility & Sales – Identify products with high visibility but low sales and potential growth opportunities.

Project Objective

The main objective of this project is to analyze Blinkit's sales and operational data using Power BI and generate meaningful, data-driven insights.

Key Objectives:-

- Analyze overall sales performance and revenue trends.
- Identify top-performing and underperforming products.
- Evaluate sales performance across different outlet types, sizes, and locations.
- Analyze product visibility and its relationship with sales.
- Understand customer ratings and satisfaction patterns.
- Identify opportunities to improve sales and outlet performance.
- Build an interactive Power BI dashboard to support data-driven decision-making.

Dataset Overview & Data Source

Dataset Overview

- Dataset: Blinkit Grocery Sales Dataset
- Domain: Quick Commerce / Grocery Retail
- Format: CSV
- Records: 8,523
- Columns: 12

Data Source

- The dataset was obtained as a publicly available dataset and used for educational and analytical purposes.
- The dataset provides information about products, sales, outlets, visibility, and customer ratings, enabling comprehensive business analysis in Power BI.

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	Item Identifier	Item Type	Outlet Establishment Year	Outlet Identifier	Outlet Location Type	Outlet Size	Outlet Type	Item Visibility	Item Weight	Sales	Rating
496	FDW27	Meat	2018	OUT027	Tier 3	Medium	Supermarket Type3	0.150122794		154.5314	4
497	FDX03	Meat	2018	OUT027	Tier 3	Medium	Supermarket Type3	0.060800116		44.4744	4
498	FDY03	Meat	2018	OUT027	Tier 3	Medium	Supermarket Type3	0.075753207		111.1202	4
499	FDG22	Snack Foods	2018	OUT027	Tier 3	Medium	Supermarket Type3	0.041180766		37.919	4
500	FDG46	Snack Foods	2018	OUT027	Tier 3	Medium	Supermarket Type3	0.032750291		112.1518	4
501	FDJ21	Snack Foods	2018	OUT027	Tier 3	Medium	Supermarket Type3	0.038341654		146.2102	4
502	FDK58	Snack Foods	2018	OUT027	Tier 3	Medium	Supermarket Type3	0.044764726		102.4016	4
503	FDN34	Snack Foods	2018	OUT027	Tier 3	Medium	Supermarket Type3	0.045542628		170.7132	4
504	FDP21	Snack Foods	2018	OUT027	Tier 3	Medium	Supermarket Type3	0.025616191		188.1872	4
505	FDR22	Snack Foods	2018	OUT027	Tier 3	Medium	Supermarket Type3	0.018472714		109.9544	4
506	FDS09	Snack Foods	2018	OUT027	Tier 3	Medium	Supermarket Type3	0.080695806		51.3008	4
507	FDS34	Snack Foods	2018	OUT027	Tier 3	Medium	Supermarket Type3	0.076387367		112.1518	4
508	FDU09	Snack Foods	2018	OUT027	Tier 3	Medium	Supermarket Type3	0.06627464		54.1956	4
509	FDU33	Snack Foods	2018	OUT027	Tier 3	Medium	Supermarket Type3	0.134057426		45.3402	4
510	FDU57	Snack Foods	2018	OUT027	Tier 3	Medium	Supermarket Type3	0.089120516		149.8708	4
511	FDU58	Snack Foods	2018	OUT027	Tier 3	Medium	Supermarket Type3	0.028871235		188.7898	4
512	FDX46	Snack Foods	2018	OUT027	Tier 3	Medium	Supermarket Type3	0.057835325		57.5562	4
513	FDX57	Snack Foods	2018	OUT027	Tier 3	Medium	Supermarket Type3	0.047037323		96.4068	4
514	FDY33	Snack Foods	2018	OUT027	Tier 3	Medium	Supermarket Type3	0.096730427		159.0262	4
515	DRY23	Soft Drinks	2018	OUT027	Tier 3	Medium	Supermarket Type3	0.108568067		42.9112	4
516	FDA11	Baking Goods	2018	OUT027	Tier 3	Medium	Supermarket Type3	0.043029436		94.7436	4
517	FDK38	Canned	2018	OUT027	Tier 3	Medium	Supermarket Type3	0.053031857		149.1734	4
518	FDO38	Canned	2018	OUT027	Tier 3	Medium	Supermarket Type3	0.072486326		78.9986	4
519	FDG32	Fruits and Vegetables	2018	OUT027	Tier 3	Medium	Supermarket Type3	0.17514326		222.3772	4
520	NCT53	Health and Hygiene	2018	OUT027	Tier 3	Medium	Supermarket Type3	0		164.5526	4
521	FDN09	Snack Foods	2018	OUT027	Tier 3	Medium	Supermarket Type3	0.034705807		241.6828	4
522	DRE13	Soft Drinks	2018	OUT027	Tier 3	Medium	Supermarket Type3	0.027570939		86.6198	4
523	FDT50	Dairy	2018	OUT027	Tier 3	Medium	Supermarket Type3	0.107714834		97.8752	4
524	FDM58	Snack Foods	2018	OUT027	Tier 3	Medium	Supermarket Type3	0		112.2544	4

Business Requirements & Key Analysis Questions

Business Requirements

- The dashboard was designed to help Blinkit management evaluate product, sales, outlet, and customer performance.

Key Analysis Questions

- What is the overall sales performance of Blinkit?
- Which item types contribute the most to total sales?
- Which outlet types and outlet sizes generate higher sales?
- How does sales performance vary across outlet locations?
- Is there a relationship between item visibility and sales?
- Which items are high-performing, moderate-performing, or low-performing?
- How do customer ratings vary across different products and outlets?
- Which areas provide opportunities to improve sales and overall business performance?

Data Understanding & Data Preparation

Data Understanding

- Reviewed the dataset structure, including tables, columns, and data types.
- Identified key product, sales, visibility, rating, and outlet-related fields.
- Classified fields into categorical and numerical variables.
- Identified key fields used to connect the tables, such as Item Key, Outlet Key, and Outlet Location Key.

Data Preparation

- Organized the dataset into relevant tables for analysis.
- Identified BlinkIT Grocery Data as the main analytical table.
- Prepared supporting tables such as Items, Outlet Info, and Outlet Location.
- Prepared the required fields for data modeling, DAX measures, KPIs, and visualizations.

Data Cleaning & Transformation using Power Query

Data Cleaning

- Promoted the first row as headers.
- Corrected data types for the required columns.
- Applied filters to remove unwanted records.
- Used Replace Values to correct inconsistent values.
- Removed duplicate records.
- Removed unnecessary columns where required**.**
- I have added the **sales per kg** by using measure.

Data Transformation

- Created a Custom Column for additional analysis (**Performance Classification**).
- Merged related queries and expanded the required fields.
- Added an additional column to the Outlet Information table (**Years of Operation**).

Outcome

- Prepared a clean and analysis-ready dataset for data modelling, DAX Calculations, KPIs, and dashboard visualization.

Data Model & Relationships

Data Model

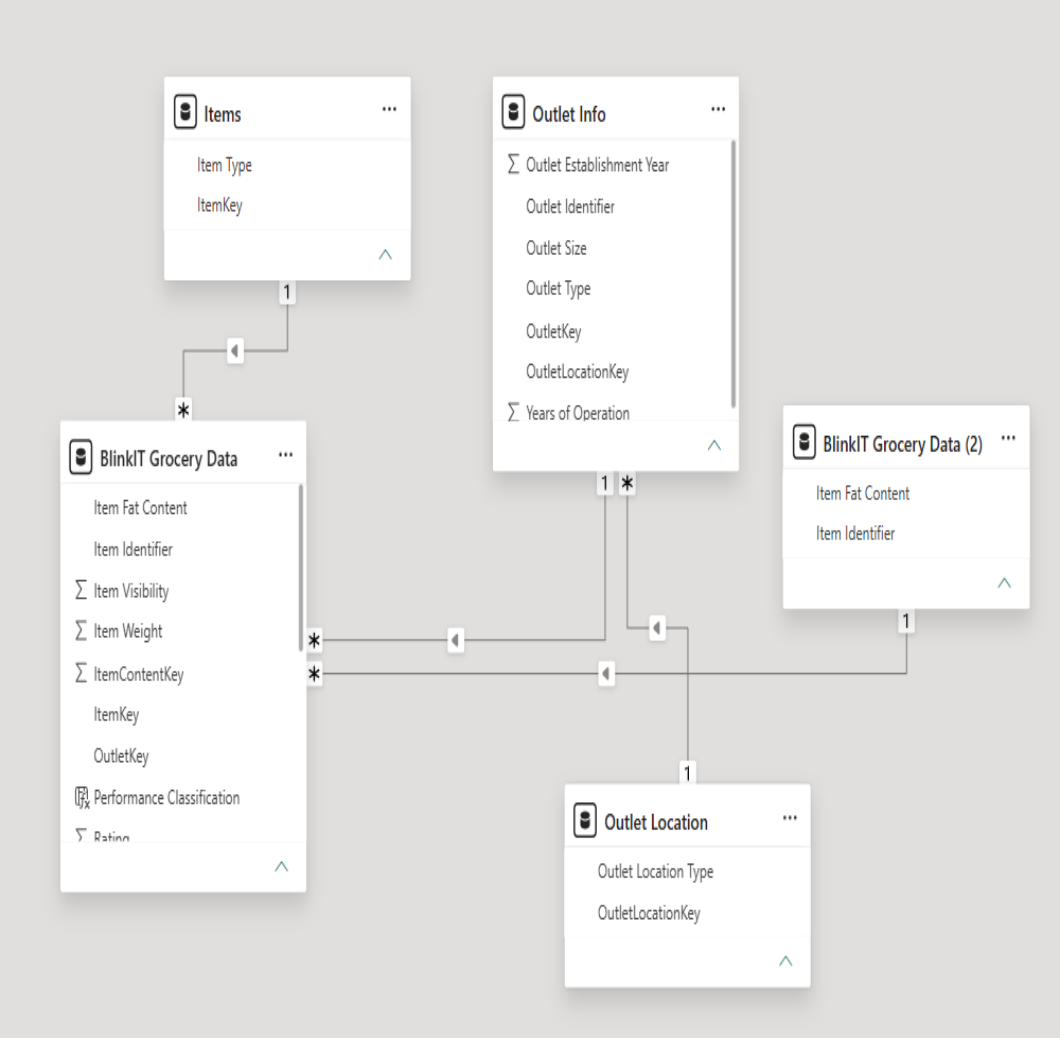
- The cleaned data was organized into related tables in power BI to support structure and consistent analysis.

Tables in the model

- BlinkIT Grocery Data – Main Analytical table
- Items – Item-related Information
- Outlet Info – Outlet Characteristics
- Outlet Location – Location-related Information
- BlinkIT Grocery Data(2) – Supporting item-related data

Relationships

- Tables are connected using relevant Item and Outlet keys.
- Relationships enable analysis across products, sales, outlets, and locations.



Data Modeling Approach

Data Modeling Approach

- Used BlinkIT Grocery Data as the main data table.
- Created separate tables for **Items, Outlet Info, and Outlet Location.**
- Connected the tables using unique identifier keys.
- Organized the data to make analysis and filtering easier.
- Used the model to create DAX measures and dashboard visuals.
- Kept **item** and **outlet** attributes separate from the main analytical data to avoid unnecessary repetition.

Simple Flow

- Main Data → Supporting Tables → Relationships → DAX → Dashboard

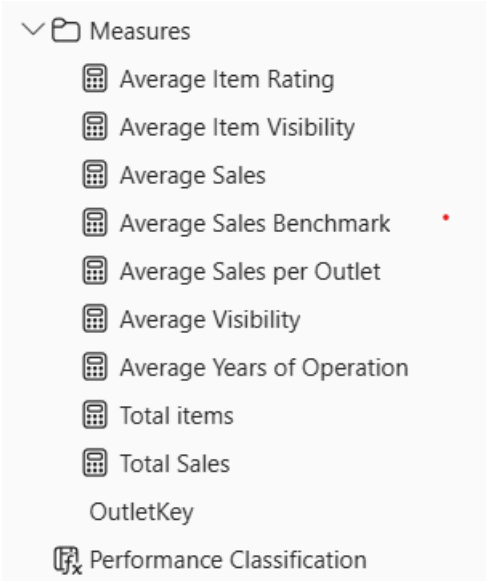
DAX Measures & Calculated Columns

Key DAX Measures

- Total Sales – Calculates overall sales generated.
- Average Sales – Calculates the average sales value.
- Average Sales per Outlet – Measures average sales contribution across outlets.
- Average Sales Benchmark – Provides a benchmark for comparing sales performance.
- Average Item Rating – Calculates the average customer rating.
- Average Item Visibility – Measures the average visibility of items.
- Average Visibility – Supports visibility-based analysis.
- Average Years of Operation – Measures the average operational age of outlets.
- Total Items – Calculates the total number of items.

Calculated Column

- A Performance Classification was created to categorize items based on their sales performance, helping identify high-performing, moderate-performing, and low-performing items.

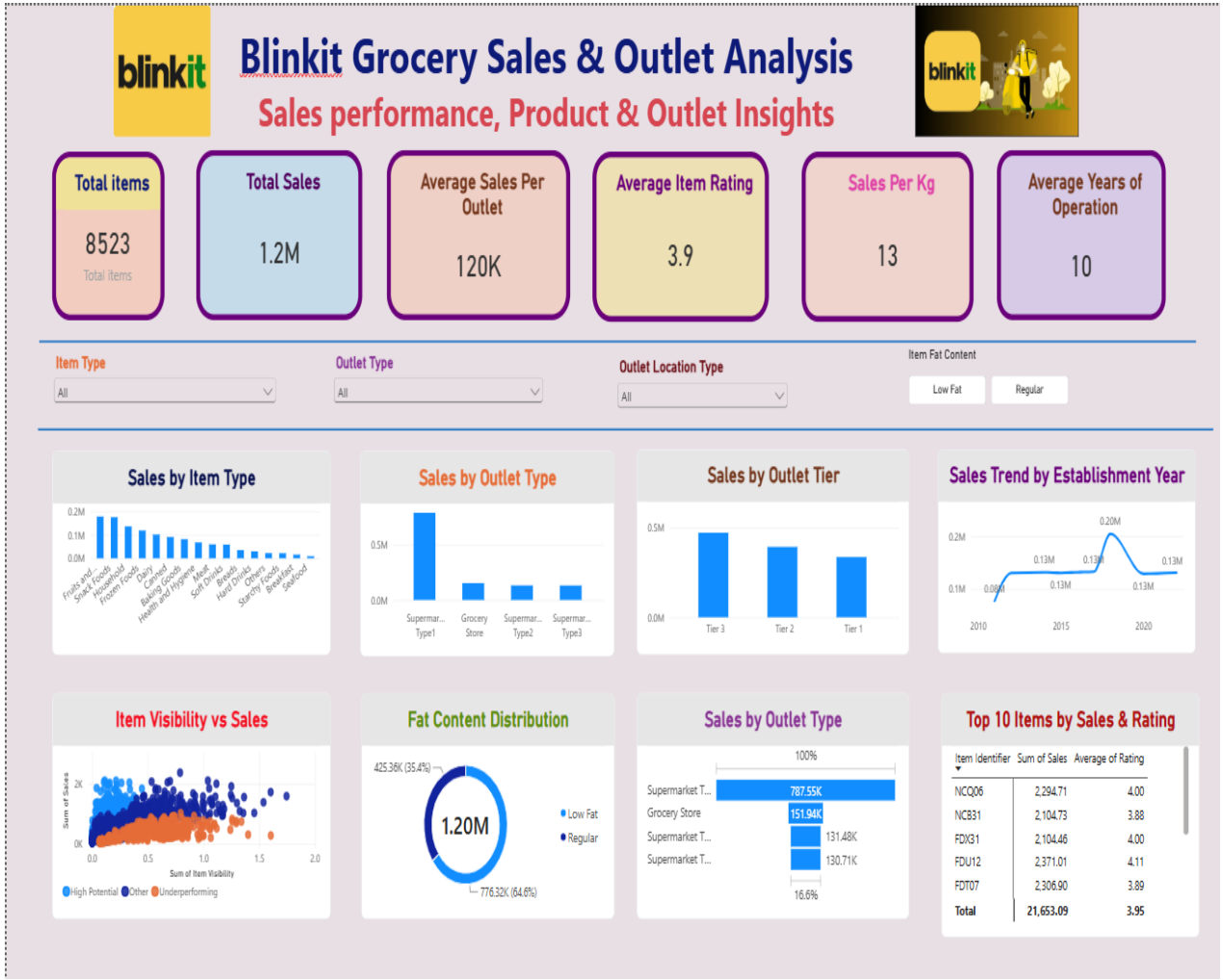


Dashboard Design & Visualizations

- The Power BI dashboard was designed as an interactive business analysis dashboard to provide a clear view of Blinkit's sales, product, outlet, and performance metrics.

Dashboard Components

- **KPI Cards** – Display key metrics such as Total Sales, Average Sales, Average Rating, and Total Items.
- **Charts & Graphs** – Used to compare sales and performance across different categories.
- **Sales Analysis** – Visualized sales by Item Type, Outlet Type, Outlet Size, and Location.
- **Visibility vs Sales Analysis** – Used a scatter chart to examine the relationship between Item Visibility and Sales.
- **Performance Classification** – Used High, Moderate, and Low performance categories to compare item performance.
- **Slicers & Filters** – Allow users to interactively filter the dashboard and explore specific segments.
- **Drill-Through Navigation** – Enables users to move from summary-level analysis to more detailed information.



Key Business Insights & Findings

- **Overall Sales:** 1.2M total sales generated across 8,523 items.
- **Product Performance:** Fruits & Vegetables and Snack Foods are major sales contributors.
- **Outlet Performance:** Supermarket Type 1 and Tier 3 outlets show the strongest sales performance.
- **Location Performance:** Tier 3 outlets contribute the highest sales.
- **Sales Trend:** Sales peaked around 2018 with fluctuations in subsequent years.
- **Customer Satisfaction:** Average item rating of 3.9/5 indicates good customer satisfaction.
- **Item Visibility**: Visibility does not always translate into sales, helping identify high-potential and underperforming items.
- **Product Mix:** Regular-fat products account for the larger share of sales.
- **Outlet Efficiency:** Average sales per outlet is around 120k.

Sales Performance:

- Fruits & vegetables and Snack foods are among the stronger performing item categories, while several other categories contribute comparatively lower sales.



Product Performance:

- Higher visibility does not always lead to higher sales. The classification identifies high-potential and underperforming items, helping target products for better promotion and placement

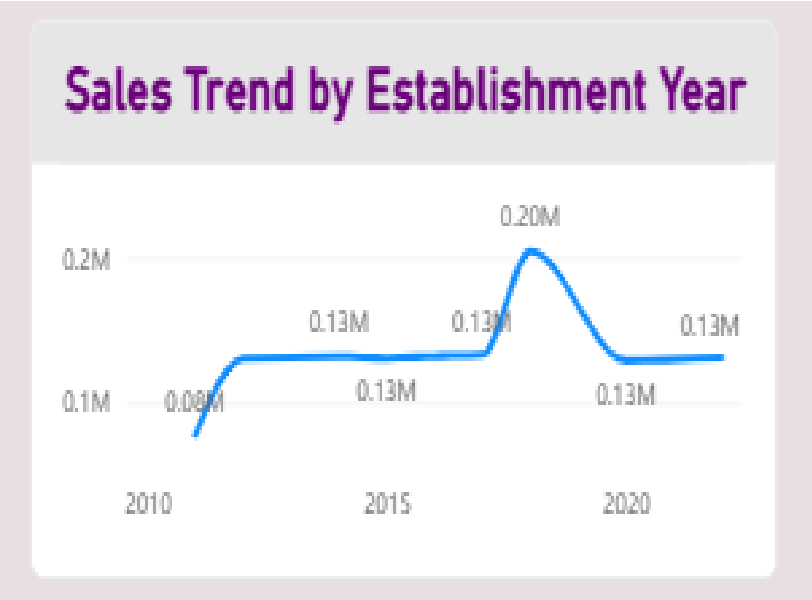


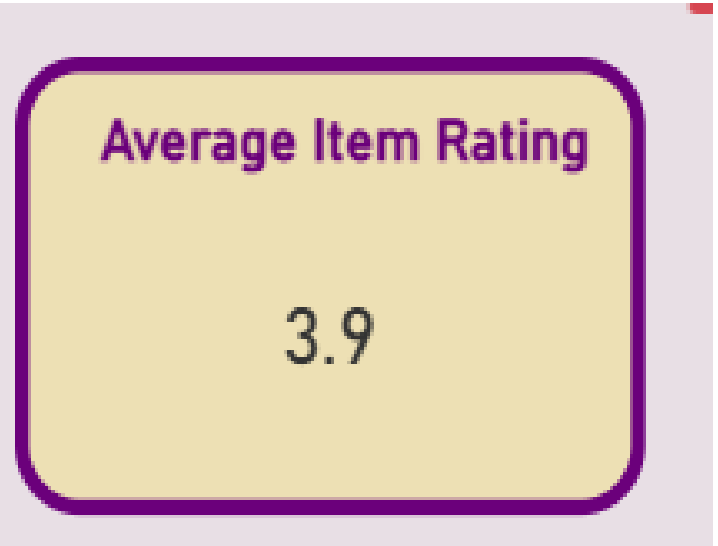
Sales Trend:

- Sales generally increased over the years, reaching a peak around 2018, followed by a decline in later years. This indicates the need to understand the factors behind the sales fluctuations.

Customer Satisfaction:

- The average item rating is around 3.9/5, indicating generally positive customer satisfaction, while individual product ratings can help identify items needing improvement.





Conclusion & Recommendations

Conclusion

- The Power BI dashboard provides a comprehensive view of Blinkit's sales, product, outlet, visibility, and customer-rating performance.
- The analysis helps identify high-performing and underperforming products and outlets.
- Comparing item visibility with sales helps highlight potential areas for improvement.
- The dashboard enables management to explore performance through interactive filters and visualizations.
- Overall, the project demonstrates how data-driven analysis can support better business decisions.

Recommendations

- Focus on high-performing products and ensure their availability across suitable outlets.
- Investigate high-visibility but low-sales products to understand why visibility is not converting into sales.
- Improve strategies for underperforming products and outlet segments.
- Use outlet-level performance analysis to identify opportunities for better resource allocation.
- Monitor customer ratings and use the feedback patterns to improve product and customer experience.
- Regularly track the dashboard KPIs to identify changes in sales and performance over time.

Challenges Faced During the Project

- **Data Quality:** Handling duplicates, inconsistent values, and data types.
- **Data Preparation:** Organizing multiple tables for analysis.
- **Relationships:** Creating correct relationships between related tables.
- **DAX Measures:** Creating and validating required calculations.
- **Dashboard Design:** Selecting suitable visuals and presenting insights clearly.
- **Performance Classification:** Defining meaningful performance categories.

Future Improvements

- **Real-Time Data:** Integrate regularly updated sales and outlet data.
- **Advanced Analytics:** Apply predictive analytics to forecast future sales.
- **Customer Analysis:** Include detailed customer behavior and purchase patterns.
- **Geographical Analysis:** Add location-based analysis to identify high-potential areas.
- **Performance Monitoring:** Create automated KPI tracking and alerts.
- **Dashboard Enhancement:** Add more interactive drill-downs and advanced visualizations.

Question & Answers

- I Welcome your Questions

THANK YOU

